Short-Shift Window Multifractal Analysis

Comparison of step $s = 1$ and $s = 7$ runs

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Scope of the packaged analysis

- Bundle analyzed: 84 runs restricted to the short-shift cases $s = 1$ and $s = 7$.
- Observables covered: counts and energy.
- Binning: 1D, 3D, and 7D.
- Window sizes present: 1095, 1670, 1825, and 3650 bins (with a few 7D families stopping at 1825).
- Diagnostic focus: windowed $\Delta h = h(0) - h(5)$, windowed $\Delta\alpha$, windowed $h(q)$, multifractal spectrum, ACF, and binned series.

Main practical point: the small-shift comparison is very clean, because the $s = 7$ trajectories are exactly a subsampling of the $s = 1$ trajectories at the common window centers.

Key result on the role of step size

What changes when going from $s = 1$ to $s = 7$?

At fixed series, binning, threshold, and window size, the values of Δh and $\Delta\alpha$ at the common timestamps are **identical** for the $s = 1$ and $s = 7$ runs. The difference is therefore not in the estimated multifractal quantity itself, but in the **sampling density of the trajectory**.

- $s = 1$: maximal temporal resolution, smoother-looking curves, stronger redundancy due to heavy overlap.
- $s = 7$: lighter and easier to visualize, but preserves the same envelope and the same extrema at shared centers.
- Interpretation consequence: physically meaningful changes should be inferred from the **shape and level** of the trajectories, not from the mere fact that $s = 1$ appears more detailed.

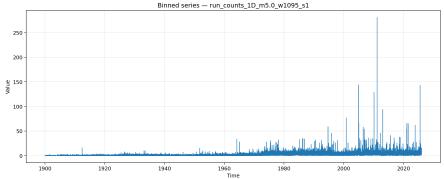
We: use $s = 7$ for clean presentation slides and $s = 1$ when one wants to inspect regime boundaries in finer temporal detail.

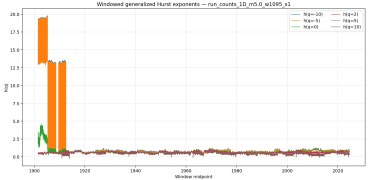
Global patterns across the 84 packaged runs

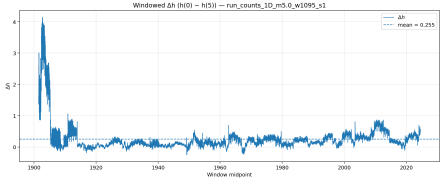
Series	Bin	$\langle \Delta h \rangle$	$\langle \Delta \alpha \rangle$	full $\Delta\alpha$	$H = h(2)$
counts	1D	0.224	0.862	14.213	0.695
counts	3D	0.150	0.546	2.080	0.795
counts	7D	0.124	0.530	0.938	0.911
energy	1D	0.847	2.224	16.641	0.489
energy	3D	0.532	1.348	3.140	0.467
energy	7D	0.460	1.118	0.922	0.521

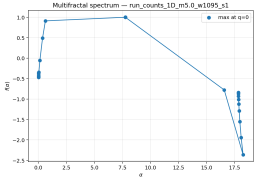
- **Energy** is systematically more multifractal than **counts**, especially in 1D.
- Coarser binning strongly compresses multifractal width, especially for counts.
- For counts, the increase of $H = h(2)$ toward 7D suggests a smoother and more persistent aggregated signal.
- For energy, multifractality remains strong even after aggregation, so intermittency survives bin coarsening more robustly.

Representative case I: counts, 1D, $m \geq 5.0$, $w = 1095$





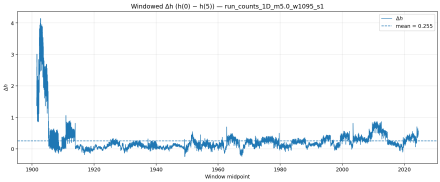


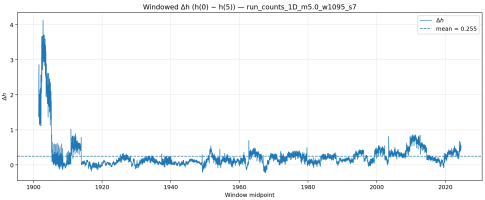


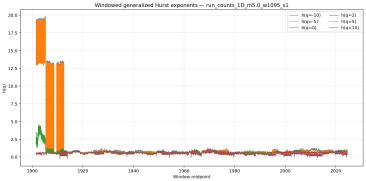
Interpretation:

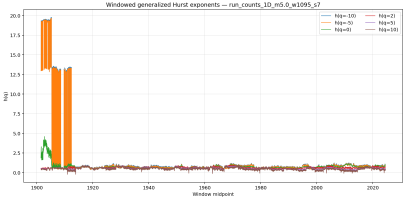
- The windowed Δh curve is highly nonstationary, with clear peaks and troughs.
- When Δh rises, the $h(q)$ fan opens up: negative-q and positive-q sectors separate more strongly, meaning that small and large fluctuations are being organized differently.
- The spectrum is broad and asymmetric, consistent with genuine multifractal heterogeneity rather than a nearly monofractal signal.

Step-size comparison in the same counts example





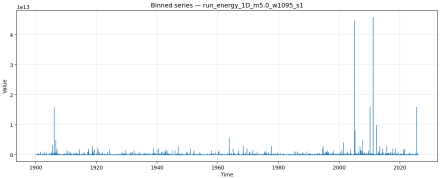


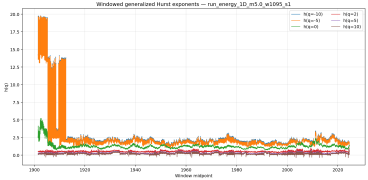


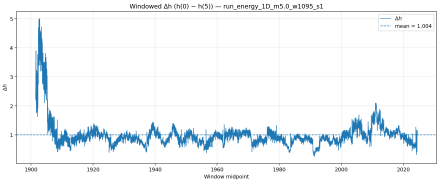
Take-home message:

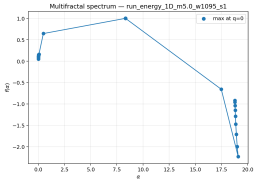
- $s = 7$ keeps the same multifractal state at shared centers.
- The visual simplification comes from less overlap, not from weaker multifractality.

Representative case II: energy, 1D, $m \geq 5.0$, $w = 1095$









Interpretation:

- Energy shows much larger Δh and $\Delta\alpha$ than counts.
- This points to stronger intermittency and a clearer separation between weak- and strong-fluctuation scaling.

Physical reading of $\Delta h = h(0) - h(5)$

How to read this observable

Large Δh means the scaling exponents sampled near $q = 0$ and toward the positive-q sector are well separated. In practice, this points to a strong difference between the scaling organization of moderate fluctuations and that of large events.

- **Peaks in Δh**: intervals where the process becomes more heterogeneous and the hierarchy of fluctuations strengthens.
- **Dips in Δh**: intervals where the signal is closer to a uniform scaling behavior or where aggregation suppresses part of the heterogeneity.
- **Agreement with $\Delta\alpha$**: when both rise together, the evidence for enhanced multifractality is stronger.
- **Need for the binned series**: to decide whether a peak is associated with burst clustering, transition between active and quiet phases, or broad redistribution of activity.

Take-home conclusions

1. The comparison $s = 1$ versus $s = 7$ is methodologically reassuring: the multifractal estimates at common centers are unchanged.
2. Therefore, the relevant scientific variation is dominated by **series type**, **binning**, and **window size**, not by the choice between these two short shifts.
3. Energy is the most multifractal observable in the package, especially in 1D.
4. Counts become markedly less multifractal as temporal aggregation increases from 1D to 7D.
5. The most defensible statement is that short-step windowing reveals **time-varying multifractality**, while the $s = 1/s = 7$ comparison confirms that this variation is not an artifact of the stride itself.

Finally:
"We use $s = 1$ for fine tracking and $s = 7$ for cleaner presentation, but the inferred multifractal regimes are the same at shared window centers."